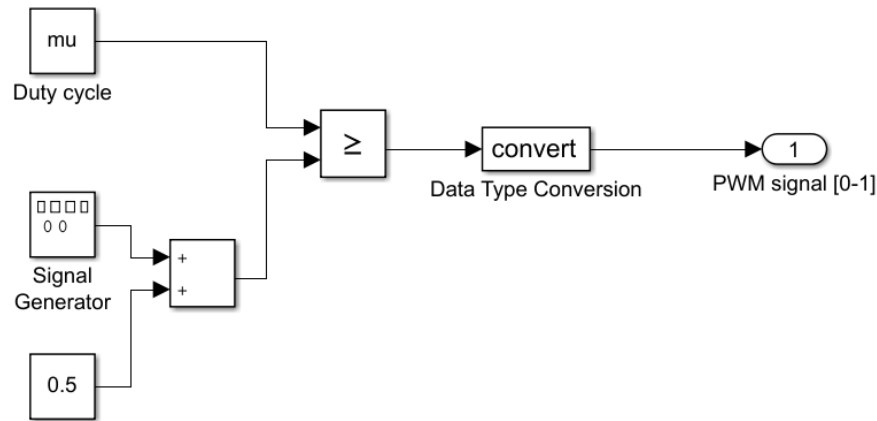


Annex 2: Simscape hints

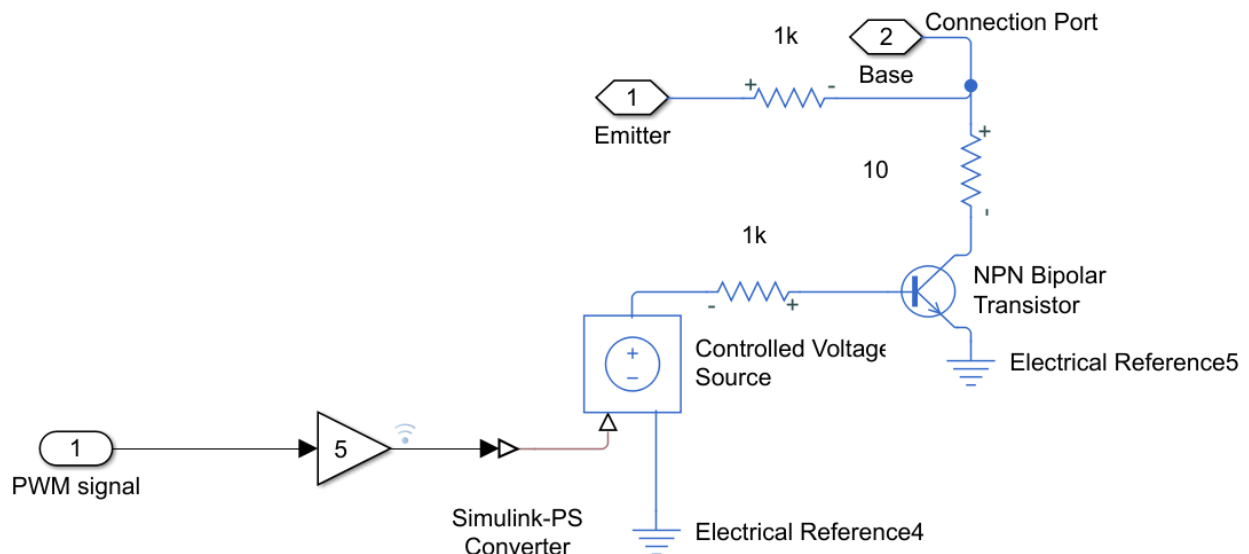
a. PWM generator



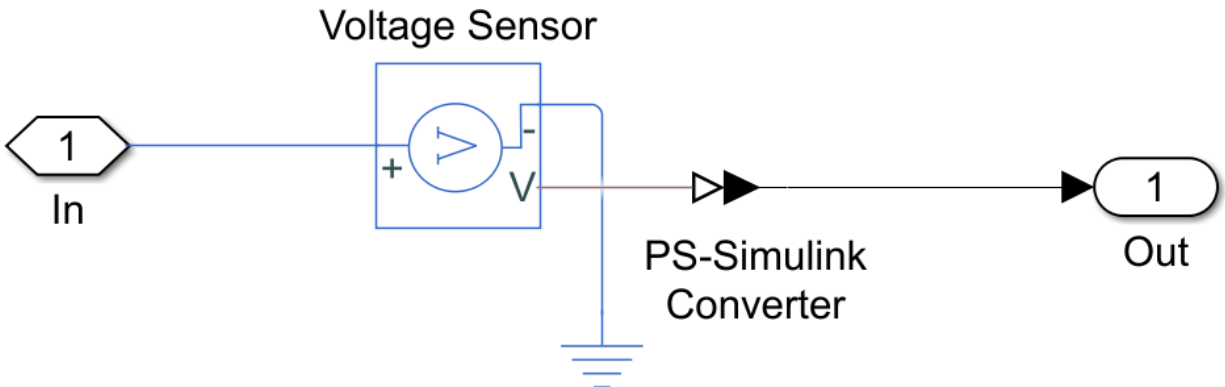
The same PWM generator topology as for normal Simulink model.

b. Transistor driver

The base of the switching transistor should be controlled by Base-Emitter current. To deactivate the transistor, we need to connect the Emitter voltage to the base with a pull-up resistor. To activate the switching transistor, we use an NPN transistor, which can be controlled by PWM signal with 5V amplitude. This signal can be generated by a microcontroller or analog circuit. In **Simulink-PS Converter** the unit of physical signal, **Volt**, should be selected.

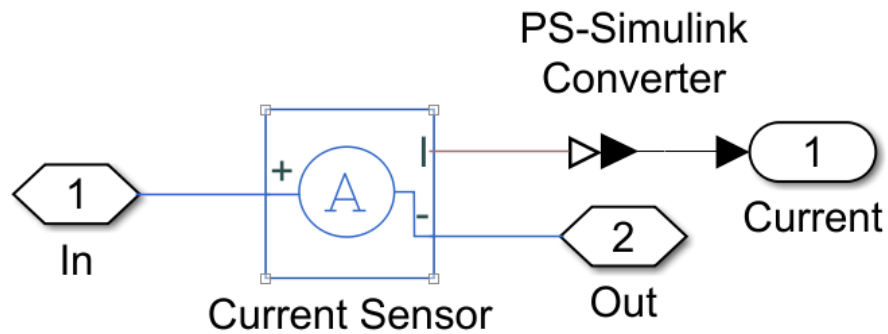


c. Voltage sensor



The voltage measurement is always referenced to the common ground. In **PS-Simulink Converter** the unit of output signal should be selected as **Volt**.

d. Current sensor



The current is measured in series with the conductor. In **PS-Simulink Converter** the unit of output signal should be selected as **Amper**.

e. Simulation step time

To open the setting window, press **Ctrl+E**.

Configuration Parameters: untitled/Configuration (Active)

Search

Solver

- Data Import/Export
- Math and Data Types
- ▶ Diagnostics
- Hardware Implementation
- Model Referencing
- Simulation Target
- ▶ Code Generation
- Simscape

Simulation time

Start time: 0.0 Stop time: 0.05

Solver selection

Type: Variable-step Solver: auto (Automatic solver selection)

▼ Solver details

Max step size: auto Relative tolerance: 1e-6

Min step size: auto Absolute tolerance: auto

Initial step size: auto ☒ Auto scale absolute tolerance

Shape preservation: Disable All

Number of consecutive min steps: 1

Zero-crossing options

Zero-crossing control: Use local settings Algorithm: Nonadaptive

Time tolerance: 10*128*eps Signal threshold: auto

Number of consecutive zero crossings: 1000

Tasking and sample time options

☐ Automatically handle rate transition for data transfer

☐ Higher priority value indicates higher task priority

OK Cancel Help Apply